CHAPTER 6 · DERIVATIONS

# GRAVITATION

3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 GRAVITATIONAL POTENTIAL OF A POINT MASS

SYLLABUS 6.3(B)

**RESULT** $V = -\frac{GM}{r}$

1. V at a point = work done per unit mass to bring a small mass from infinity to that point.
2. At distance x the attraction on a unit mass is $\frac{GM}{x^2}$ towards M, so the external force needed is $\frac{GM}{x^2}$ away from M.
3. Work done per unit mass from $\infty$ to r: $W = \int$ (from $\infty$ to r) $\frac{GM}{x^2}\,dx = [-\frac{GM}{x}]$ from $\infty$ to r.
4. $\mathbf{V = -\frac{GM}{r}}$. It is negative because the field does work as the mass moves in; V = 0 at infinity.

## 2 ΔU = MGH AS A SPECIAL CASE

SYLLABUS 6.3(D)

**RESULT** $\Delta U = \frac{GMmh}{R^2} = mgh$

1. $U = -\frac{GMm}{r}$. Raise m from the surface (r = R) to height h (r = R + h).
2. $\Delta U = -\frac{GMm}{R+h} + \frac{GMm}{R} = GMm\,h/(R(R+h))$.
3. Near the surface $h \ll R$, so $R(R+h) \approx R^2$.
4. $\Delta U \approx GMm\,\frac{h}{R^2} = \mathbf{mgh}$, since $g = \frac{GM}{R^2}$.

## 3 ESCAPE VELOCITY

SYLLABUS 6.5(B)

**RESULT** $v_e = \sqrt{\frac{2GM}{r}}$

1. Escape velocity: the minimum speed that just takes a body to infinity, where its KE and PE are both zero.
2. Conservation of energy: $\frac{1}{2}mv_e^2 + (-\frac{GMm}{r}) = 0$.
3. $\mathbf{v_e = \sqrt{\frac{2GM}{r}}}$; at the Earth's surface $v_e = \sqrt{2gR} \approx 11.2$ km s$^{-1}$. It does not depend on m.